Materials Characterization 189 (2022) 111921

Contents lists available at ScienceDirect

Materials Characterization

journal homepage: www.elsevier.com/locate/matchar

The determining role of Al addition on tribology properties and oxidation behavior at elevated temperatures of TiZrHfNb refractory high-entropy alloy

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ARTICLE INFO

Keywords:
Refractory high entropy alloys
Mechanical properties
Oxidation resistance
Wear properties

ABSTRACT

This study developed a series of $Al_x(TiZrHfNb)_{100-x}$ (x = 0, 3, 5, 7, 12 at. %) refractory high-entropy alloys (RHEAs) with an aim to improve the poor oxidation and wear resistance at elevated temperature of the base alloy TiZrHfNb. These alloys were prepared by arc melting, and the mechanical properties, oxidation and wear resistance at elevated temperatures were studied systematically. The results indicate that all the alloys are single solid solution phase with body-centered cubic (BCC) structure. The addition of Al significantly improves the strength of the RHEAs due to the large solid solution strengthening effect, and the yield strength and Al content obey the liner relation. The oxidation and wear test results indicate that the Al element will preferentially react with O atoms and accumulate on the surface to form a dense oxide layer, which prevents other elements reacting with O atoms and makes the alloy have better oxidation resistance at high temperatures, especially in alloys with 12% Al. The friction coefficient and wear rate of the RHEAs dramatically decrease with the increase of Al, which can be attributed to the combined effect of mechanical properties and oxidation resistance. This study provides an effective method for improving the oxidation resistance and wear resistance of RHEAs in a relatively high temperature range.

1. Introduction

With the development of aerospace, petrochemical engineering and other engineering fields, more stringent performance requirements are put forward for metal materials used at high temperatures [1]. The wear resistance of metal materials at high temperatures is an important manifestation of these performance requirements, which is determined by the original mechanical properties and oxidation resistance at elevated temperatures [2]. As mechanical and tribological properties generally degrade at temperatures above about 60% melting temperature, an exploration for metallic alloys based on refractory elements with high melting points may increase the maximum operating temperature [3].

High-entropy alloys (HEAs) owing to their promising mechanical properties have attracted extensive research attention [4–6]. These new alloys break the design principle with one or two primary elements and aim at producing single solid-solution phase by maximizing the entropy of mixing [7–11]. Refractory high-entropy alloys (RHEAs) are a special kind of HEAs using the refractory metals as constituents, which were proposed by Senkov, aiming at overcoming the temperature limitation of superalloy [12]. RHEA was originally composed of five refractory elements W, Nb, Mo, Ta, and V, and then expanded to nine elements (Ti, V, Cr, Zr, Nb, Mo, Hf, Ta, W) [13,14]. In addition, in order to improve the physical or mechanical properties of the alloy, Al, C, Co, Ni, Re, Si, etc. are also used as additional elements [15–20]. The most stringent problems for RHEAs are high density, poor oxidation and wear resistance at relatively high temperatures, which limit their engineering applications [21].

It is worth noting that Al possesses the strong ability to stabilize BCC phase due to its high solubility in Ti, Zr and Hf. In HfNbTaTiZr alloy, the addition of Al partly replacing Hf can increase the hardness and yield strength by 29% and 98%, respectively [17]. In $Al_xMoNbTaTiV$ alloy, the addition of Al improves the mechanical properties in a wider temperature range [22]. A small amount of Al added in MoNbTaTiV alloy can exhibit promising oxidation resistance, especially at the temperature above 800 °C for the formation of dense oxide layer [22]. In

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https://doi.org/10.1016/j.matchar.2022.111921
Received 29 November 2021; Received in revised form 16 April 2022; Accepted 19 April 2022
Available online 22 April 2022
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$Al_xTiZrHfNb$ alloys with 12 at.% Al, an (Al, Zr)-rich phase with hexagonal crystal structure was precipitated after annealing at 1000 °C and aging at 600 °C [18]. Therefore, it is realized that Al addition could bring positive effects on the microstructure and properties of such RHEAs.

The effect of Al elements on wear resistance of RHEAs at room temperature and elevated temperature has also been studied and reported. In HfTaTiVZr and TaTiVWZr alloys, the friction layer formed on the wear surface improving the bearing capacity and preventing serious deformation at 450 °C [23]. Chuang et al. investigated the wear resistance of $Al_{0.2}Co_{1.5}CrFeNi_{1.5}Ti$, and found that the wear resistance is twice that of SKH51, but the hardness is lower compared with that of SKH51, which can be attributed by its higher oxidation resistance and lower high temperature softening rate [24]. Shittu et al. compared the wear rate of AlCoCrFeNi and AISI 304 steel and Inconel 718 superalloy at temperatures above 300 °C, and found that the wear resistance of AlCoCrFeNi is significantly higher than that of AISI 304 and Inconel 718, which can be attributed to the presence of a protective oxide layer on the worn surface formed during the wear test [23]. The previous researches indicate that Al added to HEAs can improve the oxidation and wear resistance at elevated temperature. However, the relationship between tribological properties, oxidation resistance and mechanical properties over a wide range of temperature have not been studied systematically.

In the present work, TiZrHfNb with a single BCC phase, excellent plasticity, and poor oxidation resistance was chosen to be the base alloy. Five contents of Al were selected for the incorporation into base alloy for various benefits including lower density, higher strength and better oxidation resistance. The phase constitution, mechanical properties, oxidation and wear resistance of the alloy at elevated temperature were investigated. The performance and the corresponding mechanism of the alloy have been particularly concerned. This study is expected to provide more information to enrich our fundamental knowledge on friction behaviors of the RHEA and to offer a method to enhance its wear resistance through alloying with some elements such as Al.

2. Experimental procedure

2.1. Material preparation and microstructure analysis

The $Al_x(TiZrHfNb)_{100-x}$ (x = 0, 3, 5, 7, 12 at.%; donated as Al-0, Al-3%, Al-5%, Al-7% and Al-12%) alloys were prepared by arc-melting in water-cooled copper crucible under a Ti-gettered high-purity argon atmosphere. To ensure chemical homogeneity, the RHEA ingots were overturned and re-melted at least five times and then cast into a water-cooled copper mold with dimensions of 60 × 20 × 6 mm^3. The alloys were then cold-rolled by an 80% reduction in thickness, and the final thickness was 1.7–1.9 mm. After removing surface contamination by abrasive paper, the cold-rolled plates were encapsulated in evacuated argon-backfilled quartz ampoule for following heat treatment. A recrystallization annealing treatment was performed at 1000 °C for 30 min, and then quenched in water. Phase identification for these annealed RHEAs was conducted by X-ray diffraction (XRD Bruker D8) using Cu Kα radiation, scanning from 20° to 100° with a step size of 0.02°. The morphology and microstructure were characterized by the scanning electron microscope (SEM FEI Helios G4 CX), equipped with an energy dispersive spectrometer (EDS), and a Talos F200X transmission electron microscopy (TEM). The specimens for SEM observation were firstly mechanically grounded to 2000-grit SiC paper, and then electropolished for the final surface clarification in an electrolyte solution with 5 vol% perchloric acid, 30 vol% 2-Butoxyethanol and 65 vol% methanol at −15 °C. The TEM samples were prepared by electrolytic twin polishing after mechanical grounded down to ~40 μm in thickness and cut into thin discs of 3 mm in diameter.

2.2. Mechanical performance tests and oxidation resistance tests

The hardness of $Al_x(HfNbTiZr)_{100-x}$ RHEAs was tested by a Vickers Hardness tester under an applied load of 500 gf and a dwell time of 10 s, and each sample was tested at least five times. For the compression specimen, the as-cast ingots were cut into cylinders with a diameter of 5 mm and a height of 10 mm by an electrical discharge machining (EDM). Each specimen was polished to remove surface contaminations and scratched using the 2000-grit SiC paper. All the compression tests were performed at room temperature by a universal testing machine (INSTRON 3382). A constant ramp speed corresponding to a nominal strain rate of 1.5×10^{-3} s^{-1} was fixed during the compression test and the measurements were performed three times to ensure repeatability. The samples used for the oxidation resistance test were cut into a cube with a side length of 5 mm and a thickness of 2 mm by EDM. Before oxidation resistance tests, all the samples were mechanically ground by using SiC papers and thoroughly cleaned ultrasonically with ethanol. The oxidation behavior of the samples was tested on a differential scanning calorimeter (DSC, Netzsch STA 449F3). The specimens were heated from room temperature to 800 °C with a heating rate of 10 °C/ min under air atmosphere and then held at 800 °C for 6 h. During the test, the mass change of the specimens was in-situ weighed by precision balance with an accuracy of 0.0001 mg equipped in the DSC.

2.3. Room temperature and high temperature wear tests

Dry sliding wear test was performed on a ball-on-disc tribometer (RTEC MFT-5000) using Si_3N_4 ball (9.52 mm diameter) against flat samples. All the dry sliding wear tests were conducted with a fixed normal load of 5 N, reciprocating frequency of 5 Hz, the stroke length of 1 mm, sliding time of 30 min and test temperatures of room temperature (25 °C), 200 °C, 500 °C and 800 °C. The coefficient of friction (COF) was recorded in real-time by the ball-on-disc tribometer. After the dry sliding wear tests, the worn suface and the worn volume of each specimen were measured by a three-dimensional surface profiler (NPFLEX, Bruker).

In order to further analysis the corresponding wear mechanisms, the wear track of each sample after dry sliding wear tests was examined by SEM and EDS, and the oxides formed on the worn surface were analyzed by the X-ray photoelectron spectroscopy (XPS; PHI 5000 VersaProbe III with a monochromatic Al Kα X-ray source with the beam size of 100 um). Charge compensation was achieved by the dual beam charge neutralization and the binding energy was corrected by setting the binding energy of the hydrocarbon C 1 s feature to 284.8 eV.

3. Results and discussion

3.1. Microstructural characterization

In current work, the empirical parameters were summarized to predict the phase formation of HEAs, which include enthalpy of mixing (ΔH_{mix}), thermodynamic parameter (Ω), radius mismatch between constituent elements (δ) and valence electron concentration (VEC). Yang et al. proposed ΔH_{mix} and δ as standards to estimate the phase constitution, and found that the alloy is random solid solution when ΔH_{mix} and δ are in the range of −15–5 kJ/mol and 1% ~ 6.6%, respectively [25]. In addition, the above two parameters ignore the effects of enthalpy of intermetallic and mixing entropy. Thererfore, Yang et al. proposed a new parameter, Ω, which took ΔS_{conf} into account, for the evaluation of the probability to form a solid solution or intermetallics phase in multicomponent systems and found that a high value of Ω ($\Omega > 1.1$) is beneficial for the formation of solid solutions [25]. Guo et al. proposed the VEC parameter to predict the phase stability of HEAs according to the electronic structure theory [26,27]. The BCC solid solution is stable when the VEC value is lower than 6.87 according to the VEC model. These parameters can be calculated by eq. (1–4):

$$\Delta H_{mix} = \sum_{i=1, i\neq j}^{n} \Omega_{ij} c_i c_j \quad (1)$$

Table 1
Radius mismatch between constituent elements (δ), mixing enthalpy (ΔH_{mix}), the Ω parameter, and VEC for the RHEAs series.

Alloys	ΔH_{mix} (kJ/Mol)	Ω	δ (%)	VEC
Al-0%	2.500	10.748	5.733	4.250
Al-3%	−1.460	19.289	5.794	4.213
Al-5%	−3.966	7.184	5.830	4.188
Al-7%	−6.366	4.501	5.863	4.163
Al-12%	−11.898	2.398	5.929	4.100

$$\Omega = \frac{T\Delta S_{mix}}{|\Delta H_{mix}|} \tag{2}$$

$$\delta = \sqrt{\sum_{i=1}^{n} c_i(1 - r_i/\bar{r})^2} \tag{3}$$

$$\mathrm{VEC} = \sum_{i=1}^{n} c_i(VEC)_i \tag{4}$$

where, the c_i, c_j, r_i, and $\bar{r}$ represent the atomic percentages of ith and j^{th} elements, the atomic radius of ith element and the average atomic radius respectively. T is the average melting point and ΔS_{mix} is the mixing entropy. $\Delta H_{mix}$is the mixing enthalpy of the binary liquid alloy composed of the ith principal element and the jth principal element in a regular solution [28]. The specific calculation results are listed in Table 1. It can be seen that the alloys with different Al content present a stable solid solution structure, and the VEC criterion shows that each alloy has a single BCC phase structure.

Fig. 1(a) shows the XRD pattern of the $Al_x(HfNbTiZr)_{100-x}$ refractory high-entropy alloys. The results indicate that the alloys with different Al contents reveal the BCC peaks of a single solid solution, which means that the RHEAs with different Al contents possess excellent structural stability with the annealing treatment at 1000 °C. Moreover, all peaks in the XRD patterns shift to higher angles as shown in Fig. 1b and Fig. 1c. This phenomenon is caused by the change of the lattice parameters of the alloy due to the addition of Al element. Taking the (110) peak of alloy Al-12% as an example, the angle shifted from 37.004° to 37.368° and split into two peaks with similar lattice constants 3.432 Å and 3.440 Å, respectively. From the XRD results, all the RHEAs have a single BCC phase, which signifies that the Al addition does not change the original BCC crystal structure.

In order to further alalyse the element distribution and crystallographic characterization of the alloy, high-angle annular-dark field

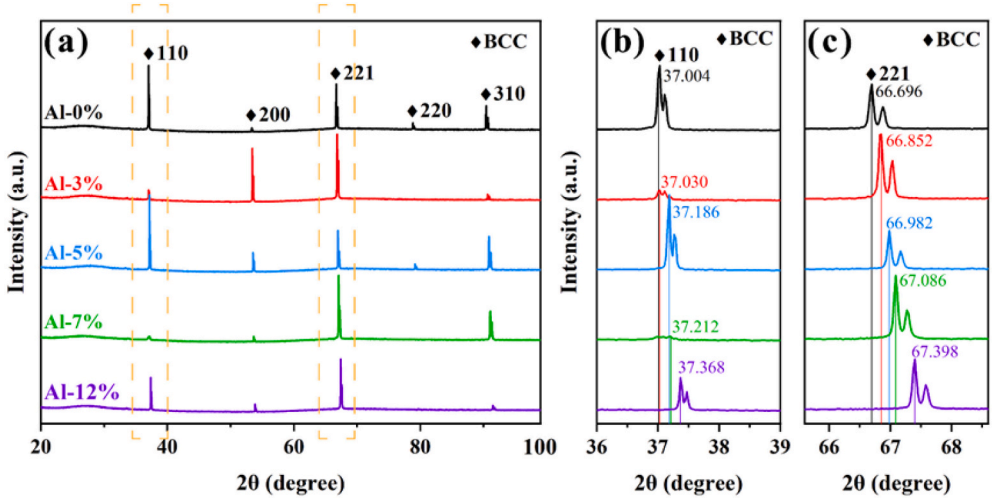


Fig. 1. (a) XRD patterns and enlarged XRD peak of each alloy: (b) (110) peak, (c) (221) peak.

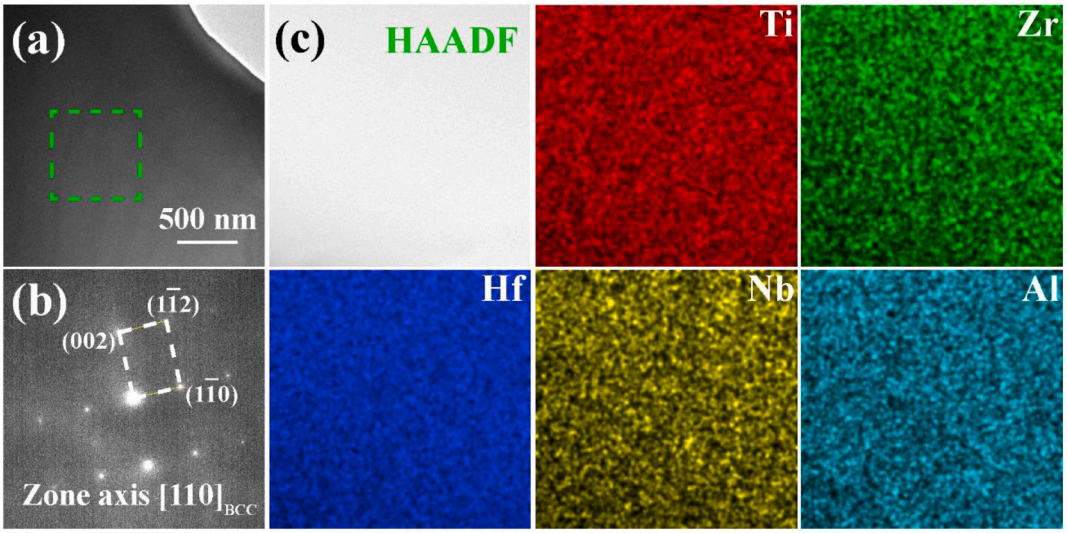


Fig. 2. (a) bright-field TEM image of the Al-12% alloy, (b) SAED pattern taken from the zone axis [110] in the Al-12% alloy, (c) HAADF-STEM image and elements mapping.

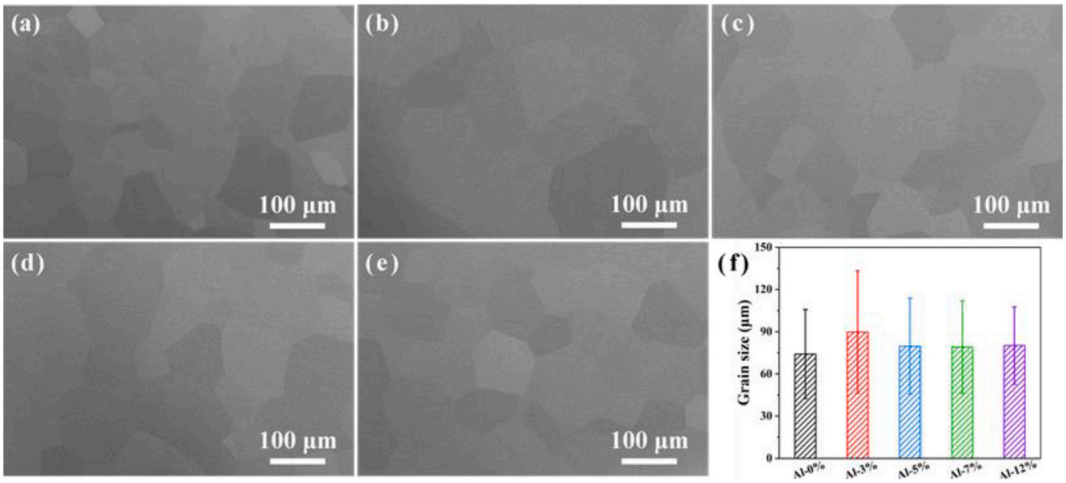


Fig. 3. SEM micrographs of recrystallized annealed alloys with different Al concentrations, (a) Al-0%, (b) Al-3%, (c) Al-5%, (d) Al-7%, (e)Al-12%, and (f) average grain size of the alloys vs. Al content.

Table 2
The theoretical density, Vickers hardness, and yield strength for the RHEAs with different Al content.

Samples	Al-0%	Al-3%	Al-5%	Al-7%	Al-12%
ρ (g/cm^3)	7.114	6.802	6.617	6.448	3.825
Hardness (HV)	272.0	304.5	319.0	344.7	389.3
Yield strength (MPa)	331.3	431.4	558.6	771.2	1094.8

scanning transmission electron microscopy (HAADF-STEM) and selected area electron diffraction (SAED) technique were used. Fig. 2 shows the bright-field TEM image, the SAED pattern, HAADF-STEM image and EDS mappings. No secondary phase is observed in the microstructure, as shown in Fig. 2c. The SAED pattern obtained in the rectangle area in Fig. 2a is shown in Fig. 2b, it is further confirmed that the structure is of a single BCC structure, and the lattice parameter is 3.433 Å based on the Bragg equation. This conclusion also confirms the results of XRD.

Fig. 3 shows SEM images of the TiZrHfNb RHEAs with different Al content after cold rolling and recrystallization annealing treatment. The microstructure of the alloy (Fig. 3a-e) consists of equiaxed grains with average sizes varying from 70 to 90 μm. Fig. 3f shows the average grain size of RHEAs with different Al content. The result indicates that the grain size is stable in different Al content alloys. Therefore, the grain refinement strengthening effect can be neglected, and the main difference in mechanical properties should be attributed to the solid solution strengthening.

3.2. Mechanical properties

The densities of the alloys can be estimated by the rule of mixtures, which related to the densities of pure elements and alloys compositions, that is:

$$\rho = \sum A_i c_i \Big/ \sum \frac{A_i c_i}{\rho_i} \tag{5}$$

where, the A_i, c_i and ρ_i represents the relative atomic mass, atomic percentages and densities of ith elements, respectively. The densities of alloys with different Al content are shown in Table 2. The result indicates density drops significantly with the increase of Al content, from 7.114 g/cm^3 (Al-0%) to 3.825 g/cm^3 (Al-12%).

The compressive stress-strain curve of different alloys at room temperature is shown in Fig. 4a. It is demonstrated that the yield strength increases remarkably with increased Al concentration. In Al-0% alloy, the yield strength is 331.3 MPa, while in Al-12% alloy, the yield strength increases by 3.3 times, reaching up to 1094.8 MPa. This is consistent with the hardness results (Al-0%, 272.0 HV; Al-3%, 304.5 HV; Al-12%, 389.3 HV), as shown in Table 2. The increase of hardness can be attributed to the significant localized lattice distortion produced by the atomic size difference and then improve the solution hardening. It is clear that Al has an atomic radius similar to that of Nb and Ti, but distinctly smaller than that of Hf and Zr. The δ for Al-0% to Al-12% alloys are 5.733, 5.794, 5.830, 5.863 and 5.929, respectively. These values are less than 6.6, indicating the present BCC phase is essentially

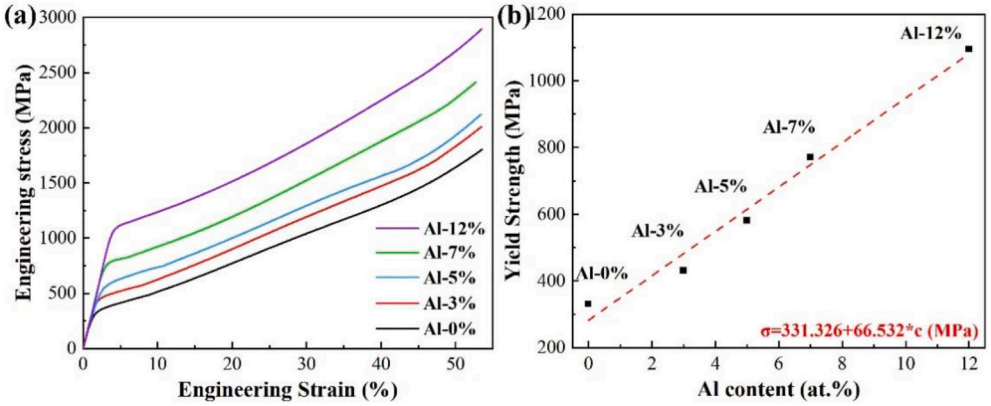


Fig. 4. (a) Stress-strain curves of the Al_x(TiZrNbHf) refractory high entropy alloys, (b) linear fit of yield strength v.s. Al concentration.

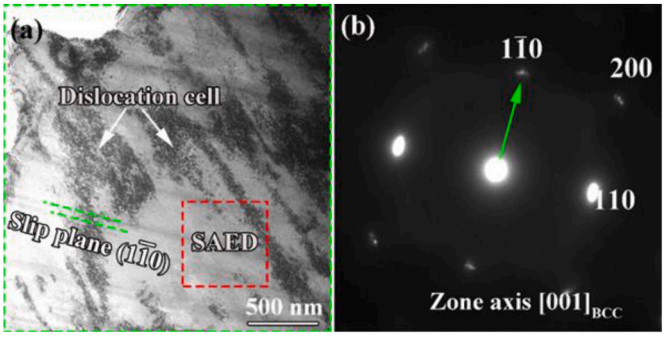


Fig. 5. (a) bright-field TEM image of the cross-section of Al-5% after compression, (b) SAED patterns of square area in (a) taken from zone axis [001].

disordered solid solution. Furthermore, the increase trend of δ agrees well with that of hardness, which is consistent with the previous report [26]. Except for lattice distortion, the distinctly strong bonds between Al and the constituent transition metals also have a strong hardening effect [16]. Therefore, the excellent hardness can be attributed to the strong lattice distortion and strong bonds between Al and the constituents. Senkov et al. have reported similar instances of solution hardening in other refractory HEAs [17].

Treating these RHEAs as pseudo-binary Al-(HfNbTiZr) alloys, the yield strength as a function of Al content was calculated to fit a linear relation, $\sigma = 331.326 + 66.532 \cdot c$ (MPa). The main difference in the yield strength of different alloys is controlled by the solid-solution effect caused by the addition of the Al. In addition, the stress-strain curve also illustrates that compressive strain has exceeded 50% in each specimen, which reveals that the plasticity of the base alloy has not been sacrificed with the increase of strength.

Fig. 5 illustrates the TEM image of the cross-section surface after compression test in the Al-5% alloy. A great number of dislocations and slip bands have been observed in the specimen, as indexed in Fig. 5(a). In order to confirm the slip plane, SAED was conducted in the rectangle area in Fig. 5(a), as shown in Fig. 5(b). The slip plane can be determined as $(1\bar{1}0)$ according to the SAED pattern.

3.3. Oxidation behavior

Before the high-temperature wear test of the RHEAs, the oxidation resistance of samples with different Al content at high temperatures are analyzed. Pre-oxidation experiments can help to establish a preliminary

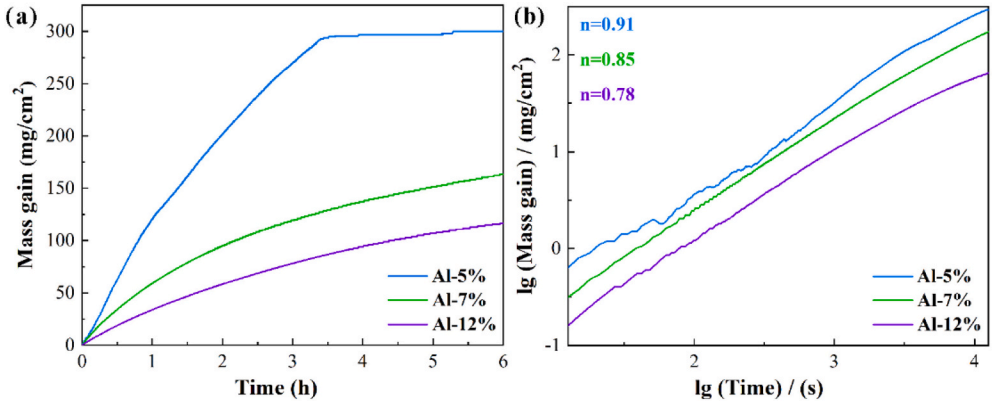


Fig. 6. (a) mass gain curves of Al-5%, 7%, and 12%; (b) double logarithmic plots of mass change against time.

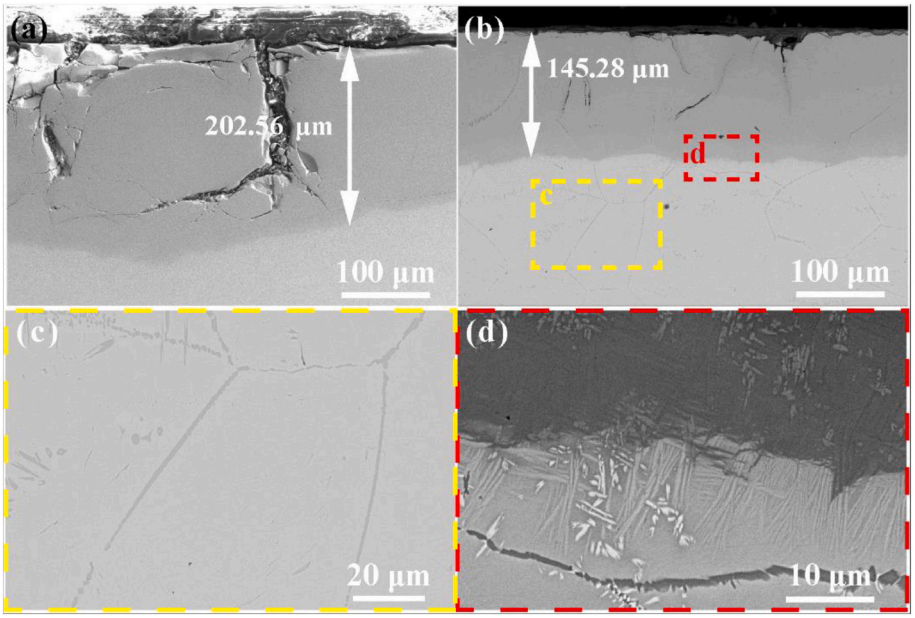


Fig. 7. SEM micrographs of the cross-section in different alloys after oxidation. (a) Al-7%, (b) Al-12%, (c) (d) magnified SEM image of c and d areas in Fib.7b.

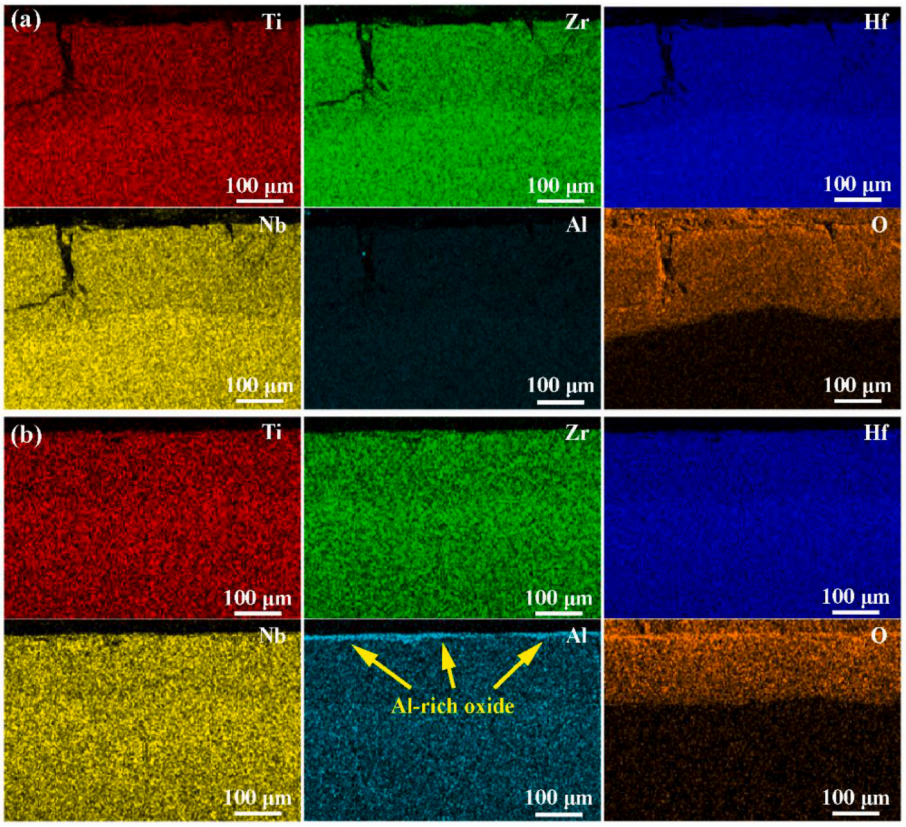


Fig. 8. EDS mapping of the cross-section in different alloy after oxidation. (a) Al-7%, (b) Al-12%.

understanding of the wear properties at elevated temperatures. Fig. 6 presents the mass gain curve of the RHEAs in the oxidation test. The data for Al-0% and Al-3% are not included in Fig. 6. Because Hf element is prone to pulverization when oxidized at high temperatures, the oxidation data of Al-0% and Al-3% cannot be measured. Fig. 6a indicates the relationship between the mass gain and oxidation time of the Al-5%, Al-7% and Al-12% alloys. It can be seen that the mass gain curve of Al-5% alloy increases with the holding time and becomes stable after roughly 3 h, which indicates that Al-5% alloy has been completely oxidized. Among the three alloys, Al-12% alloy has the least mass gain, exhibiting the strongest oxidation resistance. The results indicate that the oxidation resistance presents a positive correlation with Al addition. The oxidation behavior can be described by a power-law equation as shown in eq. (6):

$$\Delta m = k_1 t^n \tag{6}$$

where, Δm is the mass gain per unit area, t is the exposure time (s), k_1 is the oxidation rate constant, and n is the time exponent. There are two common oxidation types of metallic alloys [29]. The first type is the linear oxidation behavior, which indicates that the oxidation process is controlled by oxidation reaction rate. In this situation, the value of n in Eq. (6) is approximately 1. The second type is the parabolic oxidation behavior, which indicates the oxidation process to be diffusion controlled. In this situation, the value of n in Eq. (6) is below 0.5. The n value for Al-5%, Al-7% and Al-12% is 0.91, 0.85 and 0.78, respectively, as shown in Fig. 6b, which indicates that the oxidation type of these alloys is between parabolic and linear, and Al-12% alloy has the minimum oxidation rate. Therefore, the main wear mechanism of the alloys should be oxidative wear at elevated temperatures.

Fig. 7 shows the oxide scale formed on the Al-7% and Al-12% alloys after oxidation test. It can be seen that in Al-7% alloy the oxygen-enriched layer is roughly 202 μm and the obvious crack with large size has been observed (Fig. 7a), which indicates that the oxygen-enriched layer is porous. While in Al-12% alloy, the oxygen-enriched layer is relatively dense with the uniform thickness of 145 μm, as indexed in Fig. 7a. The enlargement view of area c and d are shown in Fig. 7c and d. It's obvious that the precipitates form on the grain

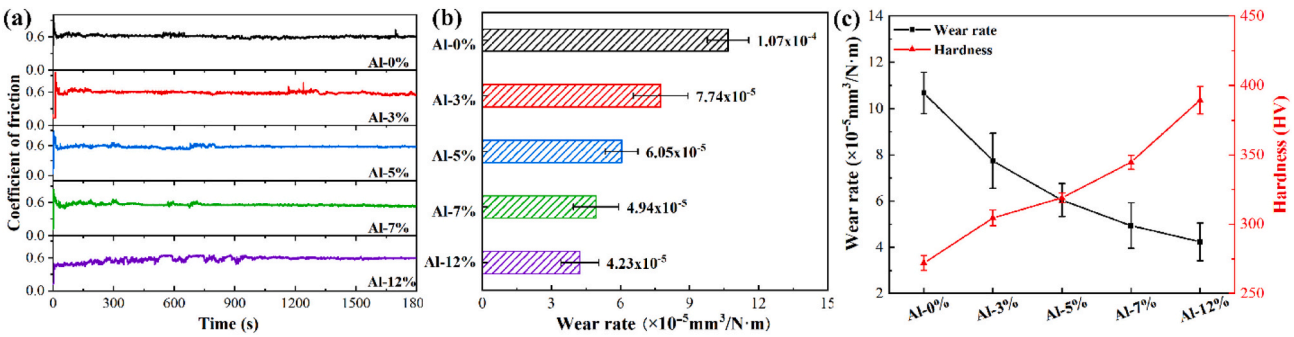


Fig. 9. (a) COF curves and (b) wear rate of RHEAs with different Al content, (c) the relationship between wear rate, hardness and Al content.

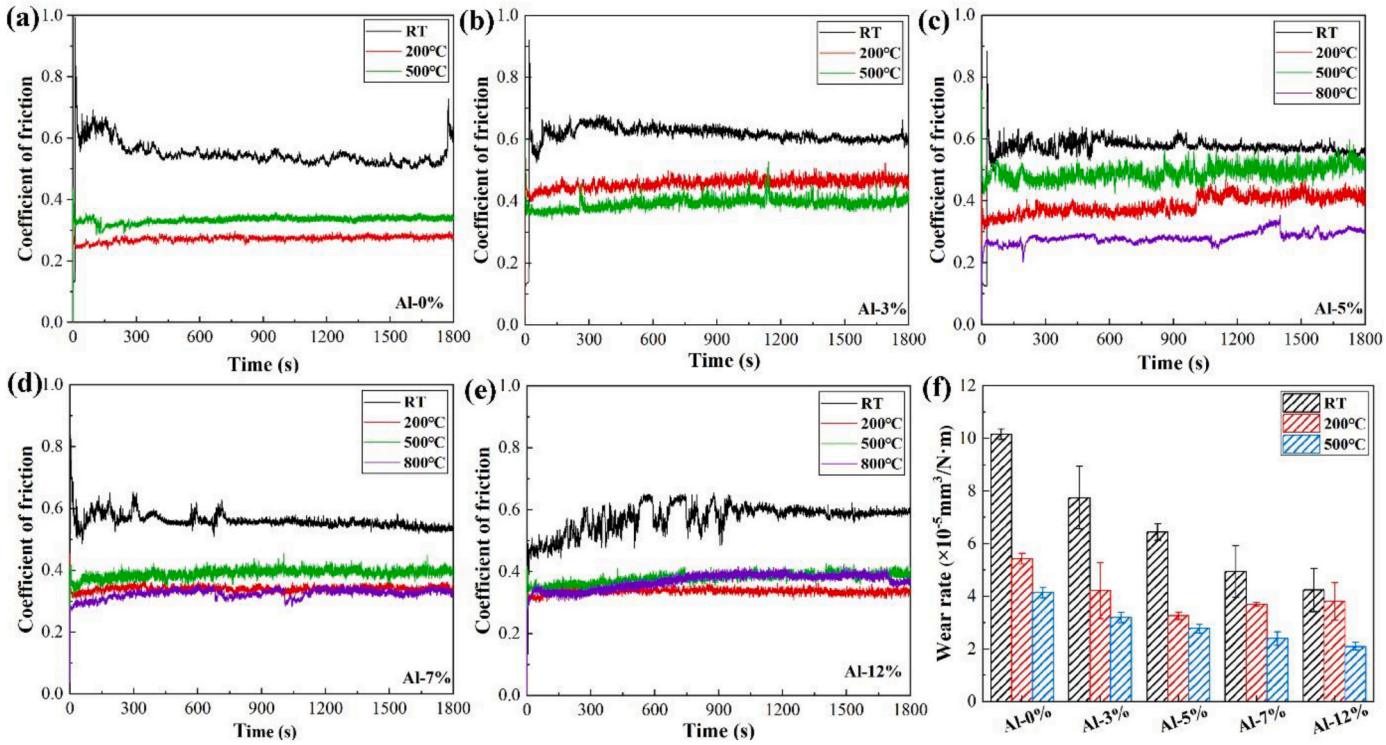


Fig. 10. (a)-(e) COF curves of five alloys at different temperatures, (f) comparison of wear rates of five alloys at room temperature, 200 °C and 500 °C.

boundary and the interface between oxygen-enriched layer and matrix. The precipitates in the interface are needle-shaped, while the precipitates in matrix is short rod-shaped. The EDS analysis (Point ID) confirms that these precipitates are enriched in Zr and Al atoms. According to previous reports, it can be determined that the precipitates are Zr_5Al_3 [18,30]. Fig. 8 shows the EDS mapping of the elements distribution (O, Ti, Zr, Hf, Nb, Al) in the cross-section of Al-7% and Al-12% specimens after oxidation test. In Al-7% alloy, a uniform oxide layer has been observed with the segregation of O atoms (Fig. 8a). While in Al-12% alloy, a stable and dense oxide layer has been observed on the surface, which could prevent the alloy from oxidation and help to improve the wear resistance of the alloys [31]. The base elements (Ti, Zr, Hf, Nb) haven't segregated in the oxide layer according to the EDS spectrum. The Al atoms form a very thin(~5 μm) oxide layer on the outermost side of the oxide layer, which inhibits the further oxidation of the alloy, thereby improving the oxidation resistance of the alloy (Fig. 8b). The formation of Al oxide layer will have a vital impact on the friction and wear properties of the alloy at elevated temperatures [32].

3.4. Friction and wear behavior

3.4.1. Room temperature coefficient of friction and wear rate

The curves of COF vs. sliding time of the alloys with different Al contents are displayed in Fig. 9a. The evaluated wear behavior is a function of normal load at reciprocating conditions. The corresponding wear rate can be calculated by eq. (7):

$$W = \frac{V}{D \cdot F} \tag{7}$$

where W is the wear rate, V is the worn volume, F is the normal load and D is the sliding distance. As shown in Fig. 9a, the average COF of each alloy tested at room temperature is in the range of 0.5–0.6, and it's relatively stable with the increase of sliding time. The wear rate of alloys with different Al contents is shown in Fig. 9b. It can be seen that the wear rate of the alloy decreases significantly with the increase of Al content. For Al-0% alloy, the wear rate is 1.07×10^{-4} mm^3 /N·m, while for Al-12% alloy, the wear rate has dropped by 60.5% (4.23×10^{-5} mm^3 /N·m) compared with that of Al-0% alloy. Fig. 9c shows the wear rate and Vickers hardness of the RHEAs with different Al contents. It is obvious that the hardness increases with Al content and the wear rate decreases with Al content. Therefore, the wear rate of the RHEAs at room temperature is determined by the inherent strength of the alloy.

3.4.2. High-temperature coefficient of friction and wear rate

Fig. 10 shows the high temperature COF and wear rate of RHEAs with different Al content. It is apparent that the COFs of the alloys decrease at elevated temperature (200 °C, 500 °C, and 800 °C) compared with that in room temperature. Due to the poor oxidation resistance of Al-0% and Al-3% alloys at 800 °C, the high-temperature friction tests at 800 °C were not carried out. The COFs of Al-5%, Al-7%, and Al-12% alloys gradually decrease with the increase of testing temperature, dropping from 0.6 ± 0.05 at room temperature to 0.3 ± 0.05 at 800 °C, which may be caused by the thermal softening effect. The COFs of these alloys at 200 °C and 500 °C basically remain at 0.4 ± 0.1. The wear rates of RHEAs tested at different temperatures are demonstrated in Fig. 10 f. It is obvious that the wear rates of these alloys gradually decrease with the increase in wear test temperature, which could be attributed to the excellent resistance to high-temperature softening and the formation of a thick oxide layer on the surface. At 200 °C, the wear rates of the alloys firstly decrease and then tend to be slightly increase with the increase of Al content. At 500 °C, the wear rates decrease with the increase of Al content. Taking Al-12% alloy as an example, the wear rate dropped by 50.6% from 4.23×10^{-5} mm^3/N·m at room temperature to 2.09×10^{-5} mm^3/N·m at 500 °C. At the same temperature (500 °C), the wear rate of Al-12% alloy has dropped by 49.5% from 4.14×10^{-5} mm^3/N·m to 2.09×10^{-5} mm^3/N·m in Al-0% alloy. The wear rates difference is related to the the wear mechanism of the alloys at different temperature.

3.4.3. Worn surface and wear mechanism

It is of great significance to study the wear mechanism of the alloys with different Al content at different tested temperatures through the surface morphology of the worn surface. Fig. 11 shows the worn surface

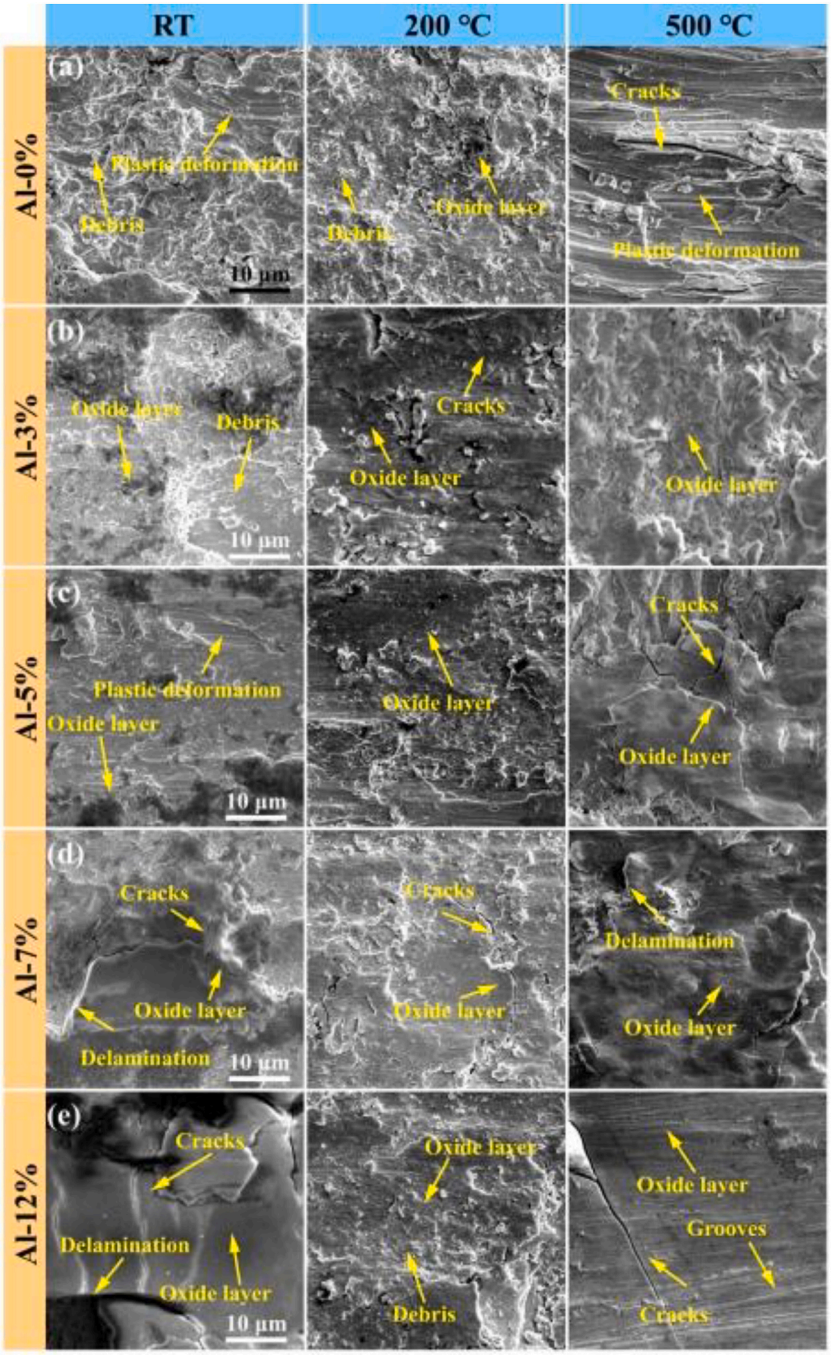


Fig. 11. SEM images of the worn surface in different RHEAs: (a) Al-0%, (b) Al-3%, (c) Al-5% (d) Al-7%, (e) Al-12%.

of five alloys tested at room temperature, 200 °C and 500 °C. The results indicate that there are a lot of large-scale wear debris and plastic deformation zones in the Al-0% alloy at room temperature for its low hardness. With the increase of Al content, more and more oxide layers appeared on the worn surface, from independent smaller lamellae to interconnected dense oxide layers. The oxidation layer is promoted by the heat produced in friction process, and a gradually dense oxide layer helps to reduce the wear rate of the alloy at room temperature [32]. The main wear mechanism transforms from general adhesive wear and abrasive wear to oxidative wear with the increase of Al content at room temperature.

The wear morphology of Al-0% alloy at 200 °C is still relatively tough, similar to that at room temperature. With the increase of Al content, a continuous oxide layer forms on the worn surface, and many cracks are generated in the oxide layer. When a critical value of the crack is reached, the accumulated internal stress in the oxidation layer could cause break of brittle oxide [33–35]. This phenomenon also confirms that the wear rate of the alloys tested at 200 °C has no obvious regularity. The wear mechanism of the alloys at 200 °C is the combination of adhesive wear, abrasive wear, and oxidative wear.

At 500 °C, Al-0% alloy has experienced severe oxidation and plastic deformation for its poor oxidation resistance. While the oxide layer in other alloys with relatively high Al content becomes more continuous and dense, especially in Al-12% alloy. A small amount of cracks in the dense oxide layer have been observed, and the density is lower compared with that at 200 °C. Therefore, oxidative wear is the main wear mechanism of the RHEAs, accompanied by a small amount of adhesive wear and fatigue wear [36].

Table 3
The element composition of the worn surface of Al-12% at different temperatures.

Temperatures (°C)	Ti	Zr	Hf	Nb	Al	O
RT	17.0	16.7	17.4	16.7	9.0	23.1
200	11.0	10.4	11.0	10.7	6.1	50.6
500	6.7	6.2	6.7	6.6	4.3	69.4

In addition, EDS analysis was performed on the wear scar of Al-12% alloy at at room temperature, 200 °C, and 500 °C (Table 3). Except for five constituent elements, O atoms are also detected in the worn surface, which are brought by oxidation that occur during the friction process. The results indicate that the O content in the alloy increases significantly with the increase of temperatures, forming more oxide layers on the worn surface. This result also confirms the main wear mechanism analyzed above. Neglecting O element, the atomic percentage of the Al element is calculated to be 11.7%, 12.3% and 14.1% at room temperature, 200 °C, and 500 °C respectively. The Al content in the oxide layer at 500 °C is largest, which corresponds to the previous oxidation resistance test, that the addition of the Al element can achieve a better anti-wear effect by forming a dense oxide layer. Fig. 12 shows the worn scar cross section morphology and the corresponding O, Al, Si and N elements distribution. In addition, the area without worn was selected for

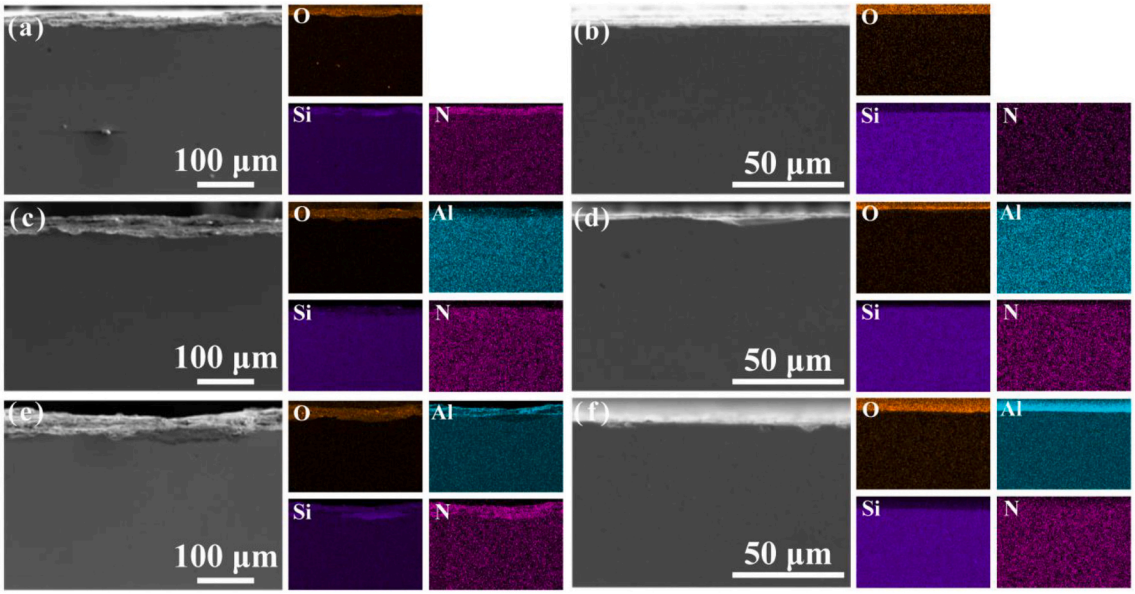


Fig. 12. EDS mapping of the wear scar cross section of the RHEAs: (a) (b) for Al-0%, (c) (d) for Al-5%, (e) (f) for Al-12%, where (b), (d) and (f) are the area outside the wear scar.

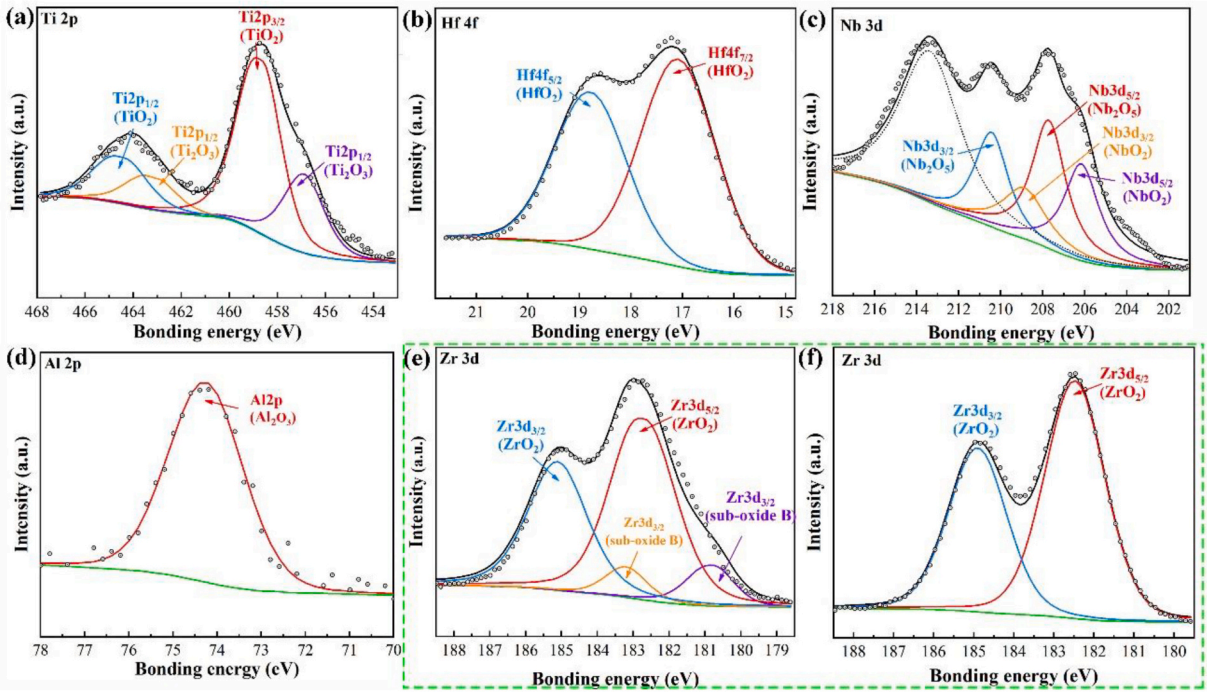


Fig. 13. XPS fitted arrow spectrum of different elements in the worn surface of Al-12% at (a-e) 500 °C and (f) 200 °C.

comparison. It can be seen that Al atoms are enriched on the surface in Al-12% alloy, compared with that in Al-5% alloy, forming a dense Al-rich oxide layer. This phenomenon appears in the worn scar and outside the worn scar, which also corresponds to the previous anti-oxidation experiment and proves that the addition of Al greatly improves the wear resistance of the alloy at elevated temperatures. Besides, a small amount of Si and N elements were also observed, indicating that slight adhesive wear occurred during the friction process.

In order to further confirm the oxidation products, XPS analysis was performed on the worn surface of Al-12% at different tested temperatures, as shown in Fig. 13. The XPS high-resolution narrow spectra of Ti 2p, Zr 3d, Hf 4f, Nb 3d and Al 2p reveal that the worn surfaces of the alloys after dry sliding wear tests in different temperatures all contain a series of oxidation products: TiO_2, Ti_2O_3, ZrO_2, HfO_2, NbO_2, Nb_2O_5 and Al_2O_3. While at 500 °C, except for ZrO_2, Zr element simultaneously forms a complicated sub-oxide B, as shown in Fig. 13e.

4. Conclusions

A series of $Al_x(TiZrHfNb)_{100-x}$(x = 0, 3, 5, 7, 12 at. %) alloys with excellent phase stability were prepared. After studying the mechanical properties and oxidation resistance of the alloys, a preliminary evaluation of their wear resistance has been made. The friction and wear behavior of the alloy at room temperature and elevated temperature and the corresponding wear mechanism have been emphatically investigated. The main findings of the study are as follows:

(1) The alloys with different Al content maintain a stable BCC single-phase structure. The equiaxed grain structure with approximate grain size is obtained by the heat treatment process. And the compressive yield strength of the alloy has been increased from 331 MPa to 1095 MPa. The oxidation resistance of the alloy with higher Al content has also been greatly improved.

(2) By improving the mechanical properties and oxidation resistance, the wear resistance of Al-12% alloy has also been greatly improved. The wear rate of the alloy at room temperature is reduced from 1.07×10^{-4} to 4.23×10^{-5} mm^3 /N·m by 60.5%. The wear rate has dropped from 4.14×10^{-5} to 2.09×10^{-5} mm^3/N·m at 500 °C.

(3) Through the morphology, EDS and XPS analyses of the worn scar, it is shown that the continuous and dense oxide layer formed at elevated temperatures, which plays an important role in improving the wear resistance, making the main wear mechanism of the alloy change from adhesive wear and abrasive wear at room temperature to oxidative wear at high temperatures.

Declaration of Competing Interest

The authors declare no conflict of interests.

Acknowledgements

This work was financially supported by the National Natural Science Foundation of China (No. 52004224) and the Fundamental Research Funds for the Central Universities (No. D5000200031) and the China Postdoctoral Science Foundation (No. 2020 M683559). We would also like to thank the Analytical & Testing Center of Northwestern Polytechnical University for their assistance with the EBSD and TEM experiments.

References

[1] V. Soni, B. Gwalani, T. Alam, S. Dasari, Y. Zheng, O.N. Senkov, D. Miracle, R. Banerjee, Phase inversion in a two-phase, BCC+B2, refractory high entropy alloy, Acta Mater. 185 (2020) 89–97.
[2] S. Alvi, F. Akhtar, High temperature tribology of CuMoTaWV high entropy alloy, Wear 426-427 (2019) 412–419.
[3] X.Q. Li, W. Li, D.L. Irving, L.K. Varga, L. Vitos, S. Schonecker, Ductile and brittle crack-tip response in equimolar refractory high-entropy alloys, Acta Mater. 189 (2020) 174–187.
[4] J.R. Morris, M.C. Troparevsky, L.J. Santodonato, E. Zarkadoula, A. Kulovits, Predicting phase behavior in high entropy and chemically complex alloys, Mater. Charact. 170 (2020).
[5] D.X. Qiao, H. Liang, S.Y. Wu, J.Y. He, Z.Q. Cao, Y.P. Lu, T.J. Li, The mechanical and oxidation properties of novel B2-ordered Ti2ZrHf0.5VNb0.5Alx refractory high-entropy alloys, Mater. Charact. 178 (2021).
[6] N.Y. Yurchenko, E.S. Panina, S.V. Zherebtsov, M.A. Tikhonovsky, G.A. Salishchev, N.D. Stepanov, Microstructure evolution of a novel low-density Ti-Cr-Nb-V refractory high entropy alloy during cold rolling and subsequent annealing, Mater. Charact. 158 (2019).
[7] Q. Li, H. Zhang, D. Li, Z. Chen, Z. Qi, The effect of configurational entropy on mechanical properties of single BCC structural refractory high-entropy alloys systems, Int. J. Refract. Met. Hard Mater. 93 (2020).
[8] Q. Zhou, Y. Du, W.C. Han, Y. Ren, H.M. Zhai, H.F. Wang, Identifying the origin of strain rate sensitivity in a high entropy bulk metallic glass, Scr. Mater. 164 (2019) 121–125.
[9] Q. Wang, J. Han, Y. Liu, Z. Zhang, C. Dong, P.K. Liaw, Coherent precipitation and stability of cuboidal nanoparticles in body-centered-cubic Al0.4Nb0.5Ta0.5TiZr0.8 refractory high entropy alloy, Scr. Mater. 190 (2021) 40–45.
[10] O.N. Senkov, J.K. Jensen, A.L. Pilchak, D.B. Miracle, H.L. Fraser, Compositional variation effects on the microstructure and properties of a refractory high-entropy superalloy AlMo0.5NbTa0.5TiZr, Mater. Des. 139 (2018) 498–511.
[11] S.A. Kube, S. Sohn, D. Uhl, A. Datye, A. Mehta, J. Schroers, Phase selection motifs in high entropy alloys revealed through combinatorial methods: large atomic size difference favors BCC over FCC, Acta Mater. 166 (2019) 677–686.
[12] O.N. Senkov, D.B. Miracle, K.J. Chaput, J.P. Couzinie, Development and exploration of refractory high entropy alloys-a review, J. Mater. Res. 33 (2018) 3092–3128.
[13] O.N. Senkov, J.M. Scott, S.V. Senkova, D.B. Miracle, C.F. Woodward, Microstructure and room temperature properties of a high-entropy TaNbHfZrTi alloy, J. Alloys Compd. 509 (2011) 6043–6048.
[14] O.N. Senkov, G.B. Wilks, J.M. Scott, D.B. Miracle, Mechanical properties of Nb25Mo25Ta25W25 and V20Nb20Mo20Ta20W20 refractory high entropy alloys, Intermetallics 19 (2011) 698–706.
[15] S. Ge, H. Fu, L. Zhang, H. Mao, H. Li, A. Wang, W. Li, H. Zhang, Effects of Al addition on the microstructures and properties of MoNbTaTiV refractory high entropy alloy, Mater. Sci. Eng. A 784 (2020).
[16] C.M. Lin, C.C. Juan, C.H. Chang, C.W. Tsai, J.W. Yeh, Effect of Al addition on mechanical properties and microstructure of refractory AlxHfNbTaTiZr alloys, J. Alloys Compd. 624 (2015) 100–107.
[17] O.N. Senkov, S.V. Senkova, C. Woodward, Effect of aluminum on the microstructure and properties of two refractory high-entropy alloys, Acta Mater. 68 (2014) 214–228.
[18] Y. Wu, J. Si, D. Lin, T. Wang, W.Y. Wang, Y. Wang, Z. Liu, X. Hui, Phase stability and mechanical properties of AlHfNbTiZr high-entropy alloys, Mater. Sci. Eng. A 724 (2018) 249–259.
[19] Y. Cao, W. Zhang, B. Liu, Y. Liu, M. Du, A. Fu, Phase decomposition behavior and its effects on mechanical properties of TiNbTa0.5ZrAl0.5 refractory high entropy alloy, J. Mater. Sci. Technol. 66 (2021) 10–20.
[20] Z. Lei, X. Liu, Y. Wu, H. Wang, S. Jiang, S. Wang, X. Hui, Y. Wu, B. Gault, P. Kontis, D. Raabe, L. Gu, Q. Zhang, H. Chen, H. Wang, J. Liu, K. An, Q. Zeng, T.G. Nieh, Z. Lu, Enhanced strength and ductility in a high-entropy alloy via ordered oxygen complexes, Nature 563 (2018) 546–550.
[21] Y.D. Wu, Y.H. Cai, T. Wang, J.J. Si, J. Zhu, Y.D. Wang, X.D. Hui, A refractory Hf25Nb25Ti25Zr25 high-entropy alloy with excellent structural stability and tensile properties, Mater. Lett. 130 (2014) 277–280.
[22] X. Yang, Z. An, Y. Zhai, X. Wang, Y. Chen, S. Mao, X. Han, Effect of Al content on the thermal oxidation behaviour of AlHfMoNbTi high-entropy alloys analysed by in situ environmental TEM, Corros. Sci. 191 (2021).
[23] M. Pole, M. Sadeghilaridjani, J. Shittu, A. Ayyagari, S. Mukherjee, High temperature wear behavior of refractory high entropy alloys based on 4-5-6 elemental palette, J. Alloys Compd. 843 (2020).
[24] M.H. Chuang, M.H. Tsai, W.R. Wang, S.J. Lin, J.W. Yeh, Microstructure and wear behavior of AlxCo1.5CrFeNi1.5Tiy high-entropy alloys, Acta Mater. 59 (2011) 6308–6317.
[25] X. Yang, Y. Zhang, Prediction of high-entropy stabilized solid-solution in multi-component alloys, Mater. Chem. Phys. 132 (2012) 233–238.
[26] S. Guo, C.T. Liu, Phase stability in high entropy alloys: formation of solid-solution phase or amorphous phase, Progress in Natural Science: Materials International 21 (2011) 433–446.
[27] S. Guo, C. Ng, J. Lu, C.T. Liu, Effect of valence electron concentration on stability of fcc or bcc phase in high entropy alloys, J. Appl. Phys. 109 (2011) 103505-1–103505-5.
[28] A. Takeuchi, A. Inoue, Classification of bulk metallic glasses by atomic size difference, heat of mixing and period of constituent elements and its application to characterization of the main alloying element, Mater. Trans. 46 (2005) 2817–2829.
[29] O.N. Senkov, S.V. Senkova, D.M. Dimiduk, C. Woodward, D.B. Miracle, Oxidation behavior of a refractory NbCrMo0.5Ta0.5TiZr alloy, J. Mater. Sci. 47 (2012) 6522–6534.
[30] L.E. Edshammar, Crystal structure investigations on the Zr-Al and Hf-Al Systems, Acta Chem. Scand. 16 (1962) 20–30.

[31] P. Zhao, J. Li, Y. Zhang, X. Li, M.M. Xia, B.G. Yuan, Wear and high-temperature oxidation resistances of AlNbTaZrx high-entropy alloys coatings fabricated on Ti6Al4V by laser cladding, J. Alloys Compd. 862 (2021).
[32] J.W. Miao, H. Liang, A.J. Zhang, J.Y. He, J.H. Meng, Y.P. Lu, Tribological behavior of an AlCoCrFeNi2.1 eutectic high entropy alloy sliding against different counterfaces, Tribol. Int. 153 (2021).
[33] E. Koraman, M. Baydoğan, S. Sayılgan, A. Kalkanlı, Dry sliding wear behaviour of Al–Fe–Si–V alloys at elevated temperatures, Wear 322-323 (2015) 101–107.
[34] M. Vardavoulias, The role of hard 2nd phases in the mild oxidational wear mechanism of high-speed steel-based materials, Wear 173 (1994) 105–114.
[35] L.W. Lan, H.J. Yang, R.P. Guo, X.J. Wang, M. Zhang, P.K. Liaw, J.W. Qiao, High-temperature sliding wear behavior of nitrided Ni45(CoCrFe) 40 (AlTi)15 high-entropy alloys, Mater. Chem. Phys. 270 (2021).
[36] L.M. Du, L.W. Lan, S. Zhu, H.J. Yang, X.H. Shi, P.K. Liaw, J.W. Qiao, Effects of temperature on the tribological behavior of Al0.25CoCrFeNi high-entropy alloy, J. Mater. Sci. Technol. 35 (2019) 917–925.